MIL-F-16895F
30 June 1987
SUPERSEDING
MIL-F-16895E
30 June 1972

MILITARY SPECIFICATION

FOOD PACKET, SURVIVAL, ABANDON SHIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the components for, and assembly of, a special purpose food packet designed for survival and to be used by Armed Forces personnel in abandon ship situations (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

EE-M-101	-	Match, Safety; and Match, Non-Safety (Strike Anywhere).
PPP-B-636	-	Boxes, Shipping, Fiberboard
QQ-A-1876	-	Aluminum Foil

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8970

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MILITARY

MIL-C-10022 - Chewing Gum.
MIL-C-10928 - Candy and Chocolate Confections.
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-882-83 - Tensile Properties of Thin Plastic Sheeting
Method B - Pendulum Weighing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency.

3.2 Food packet. The components listed in table I shall be assembled in a bag constructed as specified in 3.2.1. These components shall be arranged in the bag (see figure 1A) so that 16 food packets can be placed in the carton specified in 5.1.1 without causing crushing of the packets or preventing easy closure of the carton. The match books shall be placed against the inner walls of the bags so that the clinched ends of the match book staples do not come in contact with the walls. The filled and sealed package shall not contain excess air that causes bulging or otherwise interferes with the placement of the food packets within the carton specified in 5.1.1. (Note: The excess air may be removed manually by compressing the bag material tightly against the components prior to and during the closing operation or by removing the air mechanically to achieve a vacuum of 10 to 12 inches of mercury as registered on a gauge attached to the vacuum equipment.) The filled bag shall be closed by heat sealing with a seal not less than 3/8-inch wide. The corners and all seals of the bag shall be folded snugly against the body of the packet. The filled and sealed packets shall show no leakage when tested as specified in 4.4.1.1. The components of each packet shall not be crushed or their wrappers open, missing or torn.

TABLE I. Food packet components

Product	Number of units	Quantity of unit	Related specification and identification
Hard Candy (C)	3	1 oz.	MIL-C-10928, type V, flavor 1, style 1, class 1
Chewing gum, candy-coated	1	2 tablets	MIL-C-10022, type I, flavors 1 and 2
Matches, safety book humidity-resistant	1	20 splints	EE-M-101, type II, class 2

3.2.1 Bag construction. The bag shall be a flat style bag having inside dimensions of 3-7/8 inches in width by 5-1/2 inches in length (± 1/8 inch in either dimension) with the first dimension being the opening of the bag. The bag shall be fabricated from a material made by hot-extruding or laminating not less than 0.002-inch thick polyethylene on 0.00035-inch thick aluminum foil, conforming to QQ-A-1876 and laminating the plain side of the foil to 0.0005-inch

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thick polyester (see 4.3.1.1.1). The polyester shall be bonded to the aluminum foil by hot extruding 7 pounds per ream of clear (uncolored) polyethylene between the layers. The polyester film shall be clear and transparent so that the light reflectivity of the aluminum foil is not obscured. The combined sheet shall show no evidence of delamination when made into bags and heat sealed. The material shall be provided with sufficient slip characteristics to prevent blocking of the interior surfaces of the bag. The bag shall be sealed on 3 edges with a 3/8-inch wide heat seal (± 1/16 inch). The seams shall have a seal strength that shall be not less than 10 pounds per inch, when tested in accordance with 4.3.1.2. A tear, nick or U or V shaped notch, 1/8 inch deep (± 1/16 inch) shall be made in each side seal 1 to 1-1/4 inch from the open (unsealed) end of the bag. A 1/8 inch lip (+ 1/16 inch) shall be incorporated into the open (unsealed) end of the bag.

3.3 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Pre-acceptance inspection. The product shall be rejected if produced in plants not meeting the requirements of 3.1.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.1.1 Examination of packaging materials. Examination shall be in accordance with tables III and IV and 4.3.1.1.1. Bags for food packets shall be examined as separate lots. Sampling plans shall be as specified in table III. The acceptable quality levels (AQLs) shall be expressed as defects per one hundred units.

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TABLE II. Sampling plans for packaging materials

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
III	S-4	1 bag	bags	1.5	2.5
IV	S-3	1 bag	bags	2.5	-

TABLE III. Examination of bags for food packet

Category		Defect
Major	Minor	
101		Outer polyester film not clear and transparent.
102		Style or dimensions not as specified. 1/
	201	Width of seal (seam) less than specified at any point. 1/
103		Tear, nick or notch missing, incorrectly located or not depth specified. 1/
104		Lip missing or not depth specified. 1/
105		Tear, hole, slit or open seal (seam).
	202	Not clean.
106		Delamination.
107		Foreign odor.
108		Bag offers resistance to opening, i.e., adhesion of inner surface.

1/ Measuring device shall be calibrated in 1/32 inch increments.

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TABLE IV. Leakage and delamination of bags (see 4.4.1.1.)

Category	Defect
Major	
101	Leakage.
102	Delamination. 1/

1/ Examine after removal from water.

4.3.1.1.1 Food bags. Conformance of bag material to 3.2.1, as concerns thickness of laminae and weight of polyethylene bonding material shall be ascertained by examination of pertinent invoices, labels or other valid documents from the supplier of bag materials. Nonconformance shall indicate unacceptable bag material and use of such shall be cause for rejection of the involved bags.

4.3.1.2 Testing of food packet bags for seal strength. The bags shall be tested for seal strength in accordance with 4.4.2.1. Food packet bags shall be tested as separate lots. Test requirements shall be an average requirement. The sample unit shall be 1 bag and the sample size shall be (5) five. Results shall be reported to the nearest 0.1 pound. Nonconformance shall be cause for the rejection of the lot.

4.3.2 Examination of ration assembly. Examination shall be in accordance with tables VI through IX and 4.3.2.1 and 4.3.2.2. Sampling plans for the aforementioned tables shall be in accordance with table V. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for tables VI, VII and IX; and percent defective for table VIII.

TABLE V. Sampling plans for assembly examinations

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
VI	I	1 unsealed food packet	Food packets 1/	1.0	2.5
VII	S-4	One filled 1-piece folder box	Boxes (cartons)	0.65	2.5
VIII	S-3	1 sealed food packet	Food packets 1/	2.5	-

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TABLE V. Sampling plans for assembly examinations (cont'd)

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
IX	S-3	1 filled shipping container	Shipping containers	4.0	6.5

1/ Food packets shall be inspected as individual lots respectively.

TABLE VI. Examination of contents of food packets before sealing

Category		Defect
Major	Minor	
101		Component missing.
	201	Components of packet not arranged as specified.
102		Component crushed limiting serviceability.
103		Missing, torn or open wrapper of component.
	202	Package of food component not clean.

TABLE VII. Examination of packets in folder box

Category		Defect
Major	Minor	
101		Packet missing.
	201	Packet not arranged as specified.
102		Tear, hole, slit or open seal in packet.
103		Packet crushed limiting serviceability.
	202	Seals and corners of packet not folded as specified.
	203	Packet not clean.

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TABLE VIII. Examination of packets for leakage (see 4.4.1.1)

Category	Defect
Major	
201	Leakage.

TABLE IX. Examination of filled and closed folder boxes in shipping containers before sealing

Category		Defect
Major	Minor	
101		Folder box missing.
	201	Folder boxes not arranged as specified.
102		Markings on folder box applied by assembler missing, incorrect or illegible.
103		Closure tape not securely closing flaps of folder box.
	202	Closure tape on folder box not applied as specified.
104		Folder box crushed or otherwise damaged, adversely affecting contents.
	203	Folder box not clean.
	204	Folder box bulged.

4.3.2.1 Examination of shipping containers. Examination of shipping containers shall be in accordance with the appendix of PPP-B-636, except that the following defects shall be added to the table of defects for examination of closure and strapping: Major: Markings applied by assembler missing, incorrect or illegible.

4.3.2.2 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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4.4 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the contractor will use the procedures set forth herein. In case of dispute, as to examination or test results, the procedures specified herein will govern.

4.4.1 Examination procedures.

4.4.1.1 Leakage examination for food packet bags (Applicable to ration assembler and bag manufacturers). The filled and sealed bag shall be examined for leakage by submerging in water contained in a desiccator, or other suitable container, and maintaining a vacuum of 10 inches of mercury (atmospheric pressure of 29.9 inches) for at least 30 seconds. A steady progression of bubbles from any part of the package will indicate a leak. Isolated or static bubbles caused by entrapped air in seams or surfaces of the packages will not be considered as signs of leaks. For bag manufacturers, the bags may be filled for test purposes, with actual components as specified or with items which simulate these components.

4.4.2 Test procedures.

4.4.2.1 Determination of seam seal strength of food packet bags. Seam seal strength shall be determined in accordance with the procedure specified in Method B of ASTM, Designation 882-83. Seal specimens shall be 1/2 inch in width and shall be pulled at a uniform rate of 10 inches per minute. Readings shall record tensile strength at time of seal break.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A.

5.1.1 Level A. Sixteen food packets, arranged as specified in figure 1B, in two tiers, shall be unit packed in a one-piece folder box (see figure 2A), in accordance with style OPF, V3c of PPP-B-636, and the carton shall be closed as illustrated in figure 2B. The corrugated fiberboard material shall be B-flute. The inside dimensions of the carton shall be 14 inches in length, by 6-3/4 inches in width, by 2-1/8 inches in depth, and the maximum outside depth of the filled, closed and taped carton shall be 2-1/2 inches.

5.2 Packing. Packing shall be leve A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Eight cartons, as specified in 5.1.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Dimensions of the shipping containers shall be 14-1/2 inches in length, 7-3/16 inches in width, and 20-3/4 inches in depth. The container shall be packed flat 1 in length, 1 in width and 8 in depth. The container shall be

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strapped with a non-metallic strap or pressure-sensitive, adhesive, filament, reinforced tape, in accordance with PPP-B-636 except that one strap shall be applied girthwise around the container bisecting the 14-1/2 inch dimension. Shipping containers shall be arranged in unit loads in accor-dance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, the strapping shall be limited to non-metallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Eight cartons, as specified in 5.1.1, shall be packed as specified in 5.2.1, except that the shipping container shall be grade V3c, V3s or V4s of PPP-B-636.

5.2.3 Level C packing (for shipment of components to the ration assembler). Eight cartons, as specified in 5.1.1, arranged as specified in 5.2.1 shall be packed in a fiberboard container which is acceptable to the common carrier for safe transportation to the destination specified in shipping instructions at the lowest rate for such supplies. Shipping containers shall be in accordance with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Unit loading. When specified (see 6.2), the product packed as specified in 5.2.2 and 5.2.3 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to non-metallic strapping, except for type II class F loads.

5.4 Labeling and marking.

5.4.1 Food packets. The bags shall be legibly printed in permanent black ink with the name and address of the bag fabricator. This printing shall be placed on one face of the bag in the area adjacent to the bottom seam.

5.4.2 Carton. Cartons shall be marked as specified below and as specified in 5.4.2.1. The following information shall be legibly printed on the outside of the 14 inch by 6-3/4 inch panel of each carton with a permanent ink:

FOOD PACKET, SURVIVAL, ABANDON SHIP

Contains:
16 FOOD PACKETS

ALLOWANCE - 1 FOOD PACKET FOR 1 MAN FOR 1 DAY

TO OPEN PACKETS - TEAR ACROSS TOP AT THE NOTCH.

Retain bag to hold unused portions
Name and address of assembler
Month and year packed

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5.4.2.1 Additional marking. In addition to the information specified in 5.4.2, the following information shall be legibly printed on one side panel of each carton with a permanent ink:

> Each food packet in this carton contains food especially designed to maintain physical and mental abilities under survival conditions. It is not thirst-provoking and can be eaten in the absence of water. Use sparingly and eat only when you feel the greatest need.

5.4.3 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall me marked in accordance with MIL-STD-129.

5.4.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Intended use. This item is intended solely for survival use under abandon ship situations. It will be stored in lifeboats or liferafts, or lockers close to these craft aboard all types of ocean-going vessels.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Level of packing required (see 5.2).
c. Types and class of unit load when unit loading is required (see 5.2.1 and 5.3).

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8970-0153

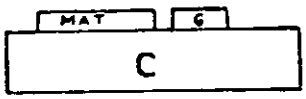
Review activities:

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Navy - MC, MS
DP - SS

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FOOD PACKET SURVIVAL ABANDON SHIP

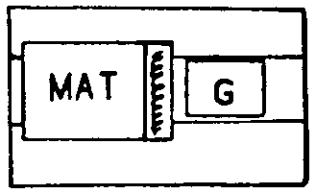
1 A. ARRANGEMENT OF COMPONENTS IN BAG



SIDE VIEW



END VIEW

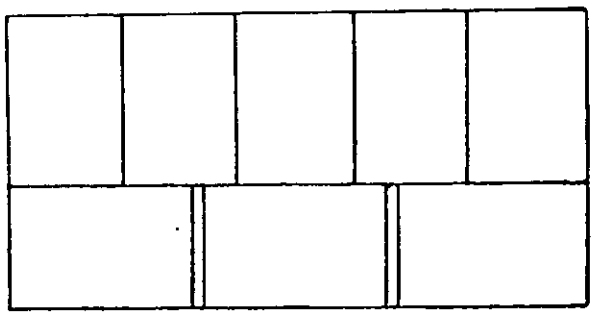


TOP VIEW

INTO BAG

G - GUM
MAT - MATCHES
C - CANDY

1 B. ARRANGEMENT OF FILLED BAGS INTO CARTON



8 PACKETS/LAYER

FIGURE 1A & 1B

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FOOD PACKET SURVIVAL ABANDON SHIP

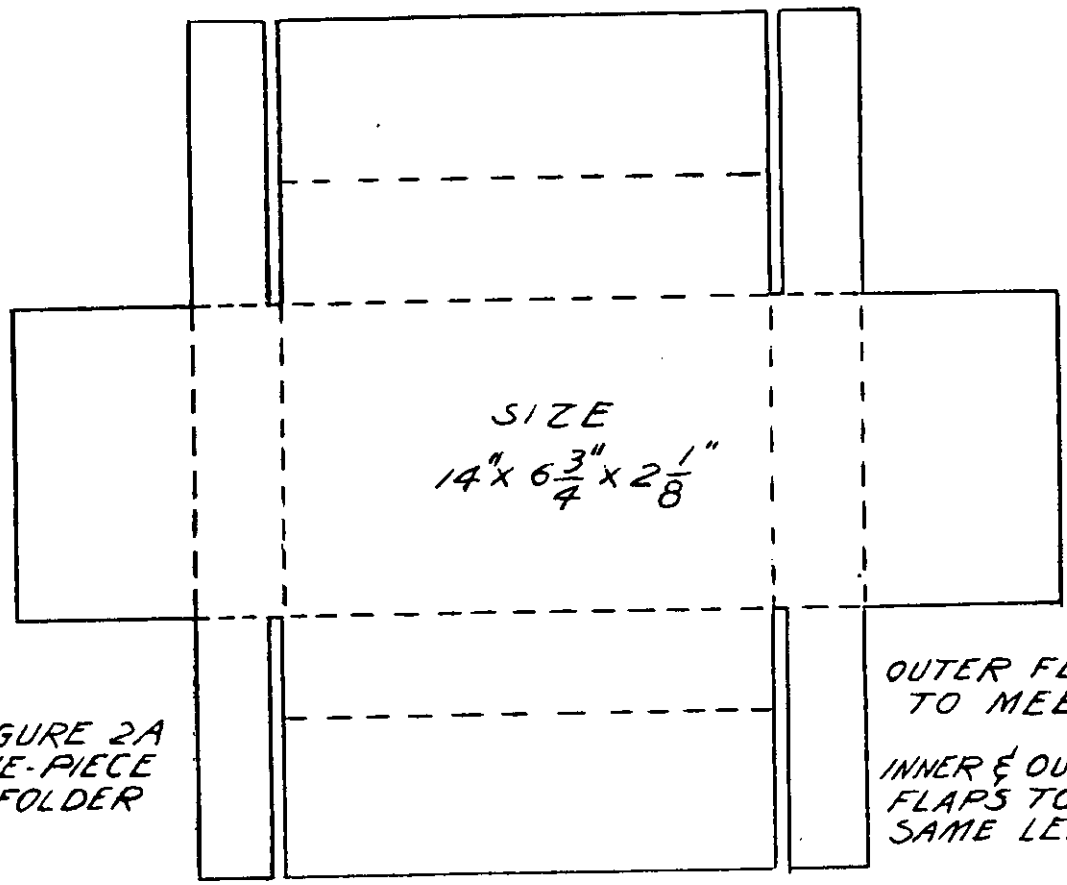
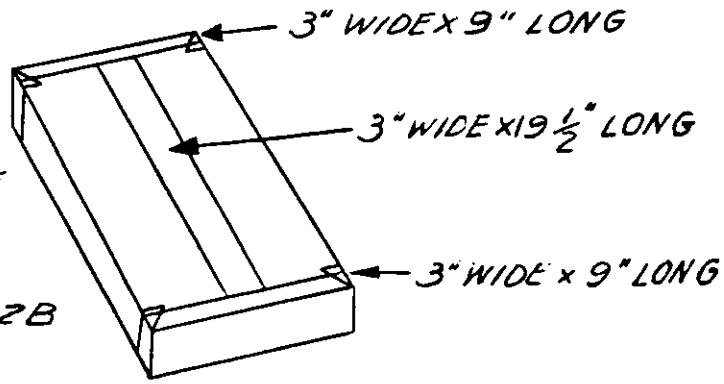


FIGURE 2A
ONE-PIECE FOLDER



TAPE CLOSURE OF ONE-PIECE FOLDER

FIGURE 2B

FIGURE 2A & 2B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-16895F	Food Packet, Survival, Abaondon Ship

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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